Generative AI – 23CSE475

Team – 04

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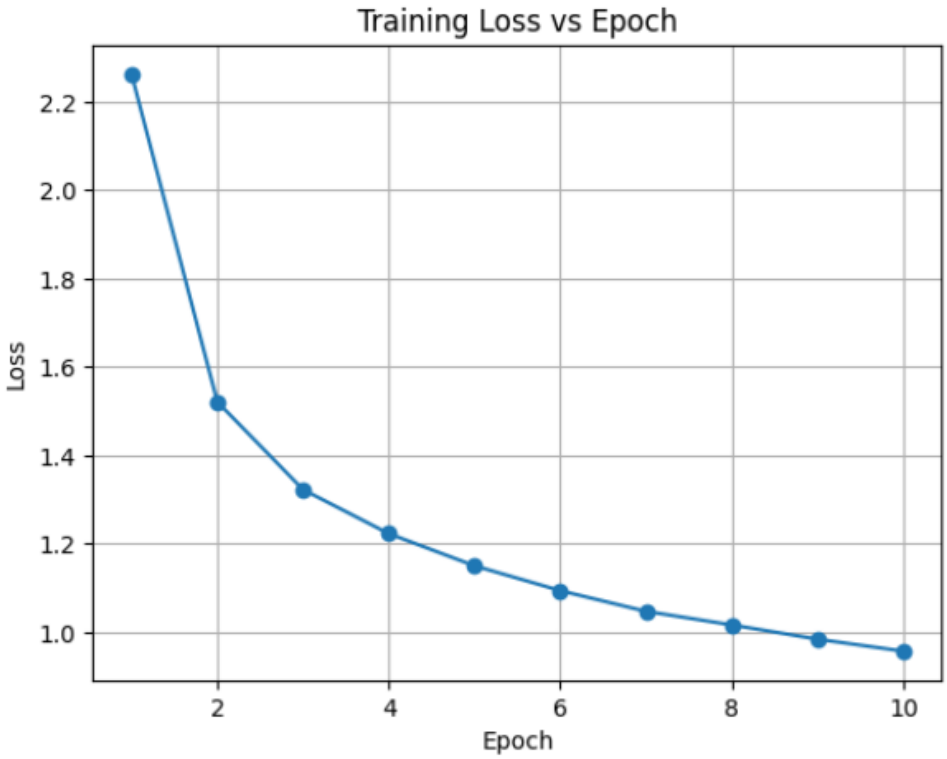
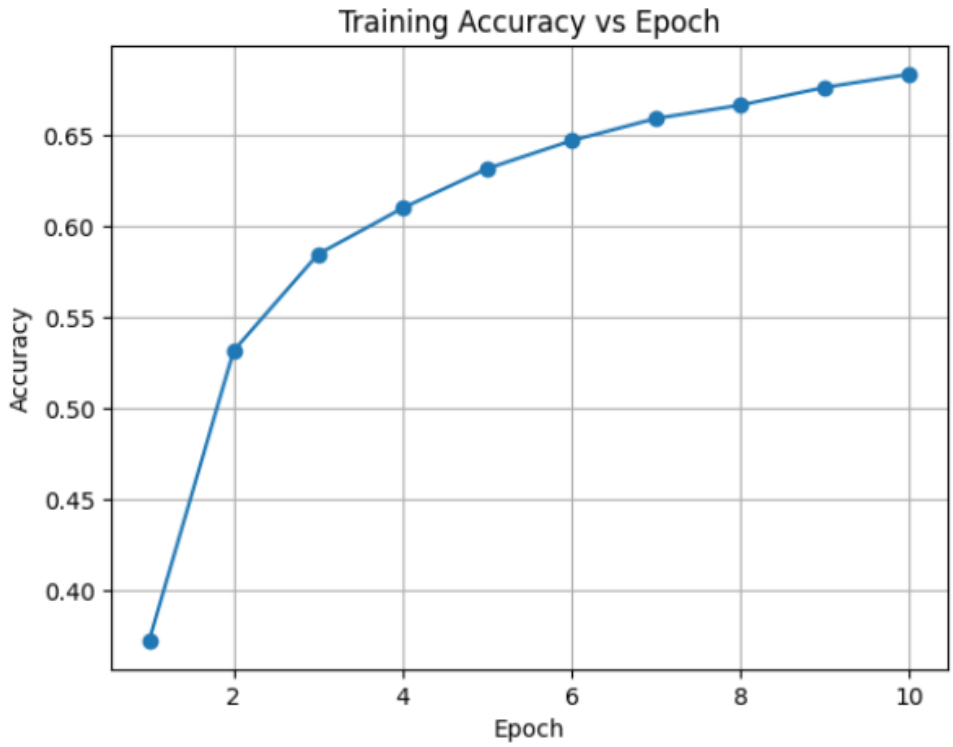
Results Quality:

Example molecules generated by the LSTM model from SMILES strings

```
Molecule 1 :  CCC(N)c1nc2nc5cc(=O)n3n(CCC4)cc33)c2)cc2c(CC13CCCCC24)CC2OC1(
Molecule 2 :  CCC(C)c1cc(Ol)c(S(=O)(=O)Nc2ccc(C(C)(O)C(O)n13)C2CC[C@H]2CCCC
Molecule 3 :  COc1ccc(C(c2ncn(-c3cccc(C(=O)NC)cc2)CC2CC(O)O)c1=O)N1SC(N)OC1
Molecule 4 :  CCNCCN(C(=O)Nc1ccn(C(=O)c2cc(O)cn(CCCNc3ccc(F)c(O)c3)N(C)c2cc
Molecule 5 :  Cc1ccccc1CCNc1cccc(C(CN2CC(C)CO)c2cccccc2n2)c1=COCCOc1ccccc1F
Molecule 6 :  Cc1cccc(C(C)(C)C)c1C(=O)N[C@@H](NC(=O)O)C(=O)NCCCCC)c1ccccc1n
Molecule 7 :  CCC1(CCC(=O)O)Sc2cc3cccc(F)c3)cc12)C(C)(C)c2cccc2)cc(-c2n[nH]
Molecule 8 :  CC(C)OCC(O)[C@@H](CC(C)(C)c1ccc2c1Cn1cc(C)c(O)c2cc(Cl=O)ccc2n
Molecule 9 :  CO[C@@H](c1nn(-c2ccccc2)c1)cc1nn[nH]n1-c1cc2c(c1)OCO3)n(C(=O)
Molecule 10 :  CC1CC(C(=O)N[C@@H](C2c3cc[n+](C[C@@H]2O)C[C@H]4C(C)(C)C)cc21)
Molecule 11 :  CC(C)Cn1c(C[C@H](C(=O)N[N+](=O)[[O-])c2ccccc2[nH]1)S(=O)(=O)[
Molecule 12 :  CCCCCCCc1c(NC(=O)Cn2ccnc6cc(C)n(Cc2ccccc2)n2cccc21)C(=O)OCC(=
Molecule 13 :  CCC(C)(C)C(=O)c1cn(OCCCCN3CCOC(N)C2)cc1CN1CCN(CCCN2CCc2nc(O)c
Molecule 14 :  CC12CCC3(CCc3ccccc3)CC1)N(C)C(=O)N(C)C)n1c(N)cc1O=C(N)COC1=O)
Molecule 15 :  CN(C)CCCC(C)(C)C(O)c1ccc(OCc2ccccc12)N=NC(CN)C(O)OSC(=1)Nc1cc
Molecule 16 :  CCCCc1nc(-[nH]c2c(=O)[nH]cn2-c(-c2ccc(OC)cc2)cc1)c1ccc(/C=C/c
Molecule 17 :  COc1ccccc1OCCCN(CCOCCCCC(O)O)CCNCCCCCNc1nnc2ncc(N(CC(=O)N)cc2
Molecule 18 :  CCCN(CCCCC[C@H](C[C@H]1CCc2ccc(OCCC(C(=O)O)NC4C[C@H]4N)c(/C=C
Molecule 19 :  CN(C(=O)n1c2c(co)nc2C(N)=O)cc1C(O)COC(=O)CCc1ccccc1-2)Oc1cc(F
Molecule 20 :  CN(c1nnc2c(CC)c(O)cc12)n1n(=O)[nH]c1OC(c1ccc(Oc2ccc(C#N)cc2)c
```

Results Analysis:





The training loss decreases steadily across epochs, indicating that the model is learning the patterns in the SMILES dataset.
The training accuracy increases over time, showing that the model is improving its ability to generate valid molecular sequences.

Quantitative Analysis:

```
Total Generated: 10
Valid Molecules: 2
Unique Molecules: 2
Novel Molecules: 2

Metrics
Validity: 0.2
Uniqueness: 1.0
Novelty: 1.0
```

$$Validity = \frac{Valid\ molecules}{Total\ generated}$$

$$Uniqueness = \frac{Unique\ molecules}{Valid\ molecules}$$

$$Novelty = \frac{Novel\ molecules}{Unique\ molecules}$$

The training accuracy increased from **0.3725 to 0.6838 over 10 epochs**, indicating that the model gradually learned the SMILES patterns. The decreasing loss and increasing accuracy show that the model improved during training.

From the generated molecules, **2 out of 10 were valid**, giving a **validity score of 0.2**. Both valid molecules were **unique and novel**, resulting in **uniqueness and novelty scores of 1.0**.